Solar Feasibility Proposal

6.0 kWp · 15 panels · 9502 kWh per year

Prepared by solarVis AI · 27 July 2026 · Reference hr1DVjA1TDLX

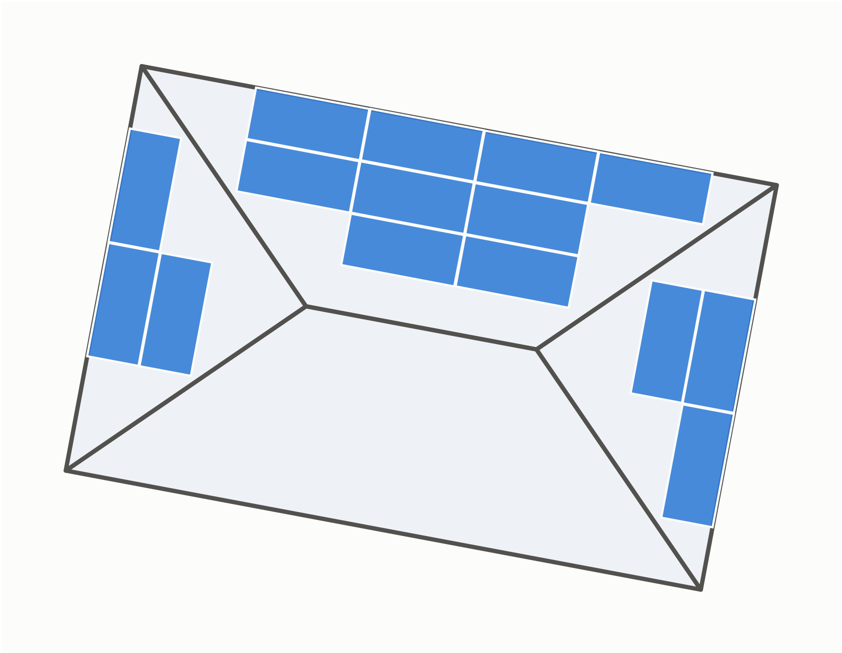
Headline results

SYSTEM SIZE	ANNUAL PRODUCTION	ANNUAL SAVING	PAYBACK
6.0 kWp	**9,502 kWh**	**€2,376**	**3.7 yr**
15 of 15 panels requested	68.9% of your usage	at €0.25/kWh	20-year net €38,721

Project inputs

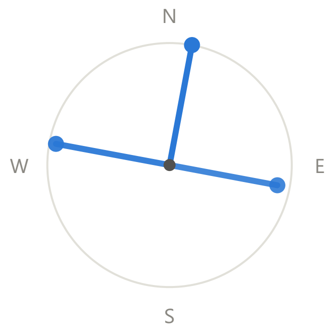
Location as entered	-34.0466, 18.4649
Location analysed	-34.046582, 18.464915
Monthly consumption	1,150 kWh
Annual consumption	13,800 kWh
Electricity unit price	€0.25 / kWh
Requested system size	6.0 kWp (15 panels)
Feasible system size	6.0 kWp (15 panels)

Roof reconstruction



Plan view of the reconstructed roof and panel layout, drawn to scale from the measured geometry.

Orientation



Ray length and weight show relative yield. This site is in the southern hemisphere, so north-facing roof faces produce most.

Facet	Azimuth	Pitch	Plan area	Sloped area	Panels
North facet	10.6°	25°	27.30 m²	30.12 m²	9
South facet	190.6°	25°	27.30 m²	30.12 m²	0
West facet	280.6°	25°	12.76 m²	14.08 m²	3
East facet	100.6°	25°	12.76 m²	14.07 m²	3
Total			**80.12 m²**	**88.40 m²**	**15**

Edge measurements

Edge	Type	Plan length	True length
eave_0	eave	11.216 m	11.216 m
eave_1	eave	7.143 m	7.143 m
eave_2	eave	11.216 m	11.216 m
eave_3	eave	7.143 m	7.143 m
hip_0	hip	5.051 m	5.319 m
hip_1	hip	5.051 m	5.318 m
hip_2	hip	5.051 m	5.318 m
hip_3	hip	5.051 m	5.318 m
ridge_0	ridge	4.073 m	4.073 m

Lengths are measured on the source satellite raster at 0.061850 m per pixel, derived from the verified Web Mercator zoom and scale. Hip lengths use true 3-D endpoint geometry; they are deliberately not the plan length divided by cos(pitch), which applies only to an edge running straight up the slope.

Energy production

Modelled monthly production across all occupied roof facets.

Facet	Panels	Capacity	PVGIS aspect	Specific yield	Annual production
North facet	9	3.6 kWp	-169.4°	1,679 kWh/kWp	6,043 kWh
West facet	3	1.2 kWp	100.6°	1,515 kWh/kWp	1,818 kWh
East facet	3	1.2 kWp	-79.4°	1,367 kWh/kWp	1,641 kWh
Total	**15**	**6.0 kWp**			**9,502 kWh**

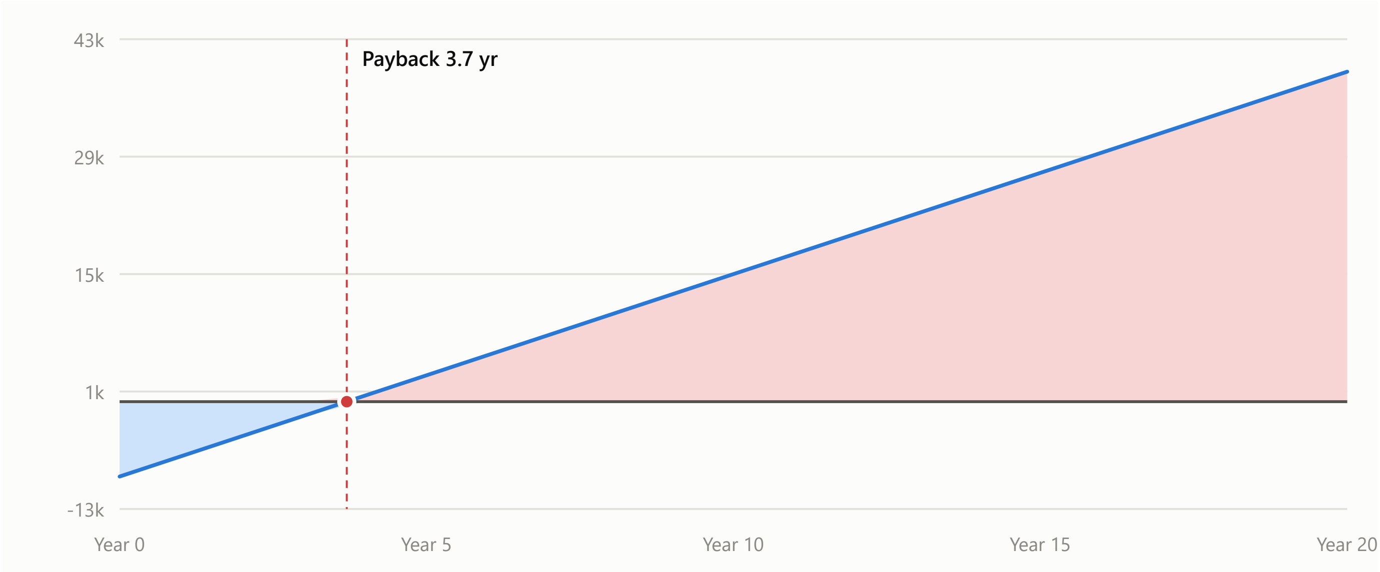
Source: PVGIS 5.3 (PVGIS-SARAH3), European Commission Joint Research Centre. One request per occupied facet, using that facet's own pitch and azimuth. Data mode: **fixture**.

Financial analysis

Annual production	9,502 kWh
Annual consumption	13,800 kWh
Energy covered (capped at consumption)	9,502 kWh
Consumption coverage	68.9%
Annual saving	€2375.55
Capital cost as quoted	$10000.00 USD
USD/EUR reference rate applied	0.87897
Rate date	2026-07-24
Rate provider	ECB via Frankfurter
Capital cost used in this analysis	€8789.70 EUR
Simple payback	3.70 years
Net benefit over 20 years	€38721.30

Reference rate source: **development fixture rate — not live market data**. The capital cost is quoted in US dollars while savings accrue in euro, so the two are never compared directly. This rate was fixed when the proposal was created and is stored with it: reopening this proposal later will not recalculate these figures at a newer rate.

Twenty-year cash flow



Cumulative position. Below the line the system is still repaying its capital cost; above it, it is returning value.

Year	Saving	Cumulative	Year	Saving	Cumulative
0	€0.00	€-8789.70	11	€2375.55	€17341.35
1	€2375.55	€-6414.15	12	€2375.55	€19716.90
2	€2375.55	€-4038.60	13	€2375.55	€22092.45
3	€2375.55	€-1663.05	14	€2375.55	€24468.00
4	€2375.55	€712.50	15	€2375.55	€26843.55
5	€2375.55	€3088.05	16	€2375.55	€29219.10
6	€2375.55	€5463.60	17	€2375.55	€31594.65
7	€2375.55	€7839.15	18	€2375.55	€33970.20
8	€2375.55	€10214.70	19	€2375.55	€36345.75
9	€2375.55	€12590.25	20	€2375.55	€38721.30
10	€2375.55	€14965.80			

Assumptions and limitations

- Fixed case property; geocoding is out of scope, so any entered location resolves to it.
- The brief prints the latitude without a minus sign, which falls in open sea. The southern-hemisphere reading is used; both values are on record.
- Uniform roof pitch of 25° across all four facets.
- Panels are 1 m x 2 m, 400 Wp, flush-mounted in the plane of the roof.
- PVGIS system loss 14%, crystSi, building-mounted.
- Electricity price held flat at €0.25/kWh.
- Savings are capped at consumption: no export compensation is assumed.
- No module degradation, inflation, maintenance cost, financing cost or tax effect.
- No shading analysis and no detection of chimneys, vents or other obstructions.
- Roof-edge setback 0 m; inter-panel gap 0.02 m.
- Imagery date is unknown and may predate recent changes to the property.
- **The USD/EUR rate in this proposal is development fixture data, not a live market rate.**
- **Production figures come from captured PVGIS fixtures, not a live call.**
- **Satellite imagery is a development fixture rendered on the verified map grid, not a live Google Static Maps request.**

This is a feasibility estimate produced from satellite imagery and modelled irradiation data, not a site survey or a binding quotation. Production and financial outcomes will vary with weather, shading, equipment, tariff changes and installation detail. A physical survey should precede any purchase.